Homework 2

Deadline: 11:59 pm September 13, 2026

Problem (1)**.** *Let S be a nonempty bounded subset of $\mathbb{R}$, and define the* diameter *of S by*

$$\operatorname{diam}(S) = \sup\{\, |s-t| : s, t \in S \,\}.$$

Prove that $\operatorname{diam}(S) = \sup S - \inf S$.

Problem (2)**.** *Prove that for every real number $a > 0$ there is a* unique *natural number n such that*

$$n - 1 \leq a < n.$$

(Your proof will need both the Archimedean Property and *the Well-Ordering Principle for $\mathbb{N}$, together with an argument for uniqueness.)*

Problem (3)**.** *Let*

$$S = \{\, x \in \mathbb{R} : x^4 - 5x^2 + 4 < 0 \,\}.$$

(a) Write S in interval notation.

(b) Find $\inf S$ *and* $\sup S$*, and state clearly whether each is attained.*

Problem (4)**.** *For each sequence below, determine whether it converges and, if so, find its limit. (You may use the limit theorems; no ε–N proof is required here, but you must justify divergence when it occurs, e.g. by exhibiting two subsequences with different limits.)*

$$\textit{(a)}\ a_n = \frac{2n^3 + n}{n^3 - 5}, \qquad \textit{(b)}\ b_n = (-1)^n \, \frac{n}{n+1}.$$

Problem (5)**.** *Give rigorous ε–N proofs of the following limit.*

$$\lim \frac{3n^2 - 1}{2n^2 + n + 5} = \frac{3}{2}.$$